

Summary of the thesis

CivicDAO: Design and Implementation of a Neighborhood DAO for the Participatory Governance of Common Goods

Andrea Bobò

1. Problem and motivation

Over the past decade, digital participatory platforms such as Decidim and Consul have shown that citizens can be involved at scale in decisions concerning the places they inhabit. These platforms collect proposals and record votes, but nothing in their operation obliges an administration to honor the result. Participation is documented, not binding.

Decentralized autonomous organizations (DAOs) are organizations whose rules are written as smart contracts on a blockchain and executed automatically, without a central authority. They offer precisely what civic platforms lack, namely execution that is binding and publicly verifiable. They also bring their problems, such as voting power proportional to capital which produces plutocratic concentration, chronic voter apathy, exposure to governance attacks and an uncertain legal status that exposes members to personal liability.

2. Research questions

Existing contributions of DAOs considered for the public sector automate a single administrative procedure or present DAO mechanisms as a generic toolbox. Neither approach confronts what is decisive for civic use: a public administration is legally responsible for the legitimacy of what is decided in its name, and therefore cannot simply be removed from the process. The promise of a DAO lies in removing exactly that kind of supervising authority. If the administration takes part in the process, it risks taking control of it and reducing citizen participation to a purely formal exercise. If it stays out, the final decision has no legal standing.

The work is guided by one question: **through which on-chain governance mechanisms can a civic DAO keep a legally responsible public body within the process as a guarantor of compliance, without that compliance control degenerating into a re-centralization that nullifies citizen participation?**

It is articulated into three sub-questions: how to ground the legitimacy of participation in residency rather than in capital (SQ1), how to keep the public body inside the process and give the community legal standing toward third parties, without granting a discretionary veto on the merit of proposals (SQ2) and how to sustain participation over time against apathy, without the perverse effects to which incentive design is prone (SQ3).

3. The design of CivicDAO

CivicDAO is a DAO operating at the scale of a neighborhood, designed to coordinate and incentivize citizen participation in the care of common goods.

Hybrid architecture. On-chain governance is paired with a third-sector entity (ETS, registered in the RUNTS) that acts as a legal wrapper. It holds legal personality, asset partitioning and represents the community toward third parties. On the other hand, the on-chain resolution is the form through which the members' assembly expresses its will. Decentralization and legal accountability therefore coexist instead of excluding one another.

Membership grounded in residency. The right to participate derives from a citizenship token that is *soulbound*, meaning permanently bound to a single address and impossible to transfer, sell or accumulate. It is issued to verified residents through national digital identity infrastructure (ANPR for residency, SPID for authentication). In this way the token-weighted logic from which plutocratic drift originates is structurally removed.

Two separate flows. In the *private flow* the community governs its own civic actions autonomously, under a strict one-person-one-vote rule, with no municipal involvement. That autonomy rests on an existing legal instrument, called the collaboration agreement. In the *public flow*, participatory budgeting takes place. Citizens allocate public funds through quadratic voting over an equal credit budget, a scheme that lets the intensity of a preference be expressed while giving every citizen the same total voice.

The bounded municipal role. Within the public flow, the municipality is confined to opening a round, fixing the slate of proposals and recording the outcome. None of these allows it to alter the scores expressed by citizens. Two complementary mechanisms make that boundary effective, because an administration can empty a participatory process in two different ways. Against an arbitrary refusal, the design introduces a codified obligation to state reasons. In this way the contract will not register the non-execution of a winning proposal unless the transaction carries the reference to a written justification, which thereby becomes immutable, attributable and time-stamped. Against silent obstruction, every transition that depends only on the passing of time is permissionless. In fact closing a round, confirming an unchallenged project, or letting an unfulfilled one expire can be triggered by anyone. The process advances without the administration. The two mechanisms interlock: inaction no longer affects the outcome, so an administration that wants to intervene must act and acting triggers the duty to justify.

Sustaining participation. Verified participation is rewarded on three planes. First, a community credit spendable only within a closed circuit of enrolled local businesses and burned on redemption, so that its value is tied to real local exchange rather than speculation. Second, a non-transferable proof of participation and third a reputation score that decays in the absence of continued engagement. Reputation unlocks functional roles but never adds voting weight, so the incentive to participate cannot turn into an accumulation of decision-making power.

Verifying actions in the physical world. A civic action such as cleaning a park cannot be observed by a smart contract. Verification is therefore optimistic, in fact the evidence submitted by the team is assumed truthful unless a citizen challenges it within a bounded window, in which case a randomly drawn jury decides.

4. Implementation and scope

The design was realized in Solidity on the OpenZeppelin libraries, developed and tested with the Foundry framework and executed on a local Anvil node. The implementation covers three of the four architectural layers: identity (`CitizenshipSBT`), governance (`ParticipatoryBudget`, `CivicProject`) and economy (`CommunityCredit`, `MerchantMembership`, `ProofOfParticipation`, `Reputation`). Within this boundary the prototype is end-to-end. A citizen can be admitted, the community can propose, vote, verify a civic action and a confirmed decision produces real economic effects. The infrastructural layer, comprising the multi-signature treasury, gasless voting, the Security Council, the legal-interface contract and a real sortition jury, was left outside the prototype. The source code including demonstrative demos is publicly available at <https://github.com/smeneguz/tesi-civic-dao>.

5. Results

Read together, these answers show that the mechanisms do not dissolve the tension between legal accountability and decentralization: they relocate it. The public body remains inside the process, but its residual power is narrowed to points that are few, explicit and verifiable.

Sub-question	Mechanism	Realized in
SQ1 Legitimacy from residency	Soulbound one-person-one-vote citizenship, equal per-voter credit in quadratic voting, electorate snapshotted on-chain	CitizenshipSBT, ParticipatoryBudget
SQ2 Public body in, no veto on merit	Separate private and public flows, codified obligation to state reasons, off-chain legal wrapper	ParticipatoryBudget, CivicProject, legal interface (designed)
SQ3 Sustained, non-perverse participation	Closed-loop credit, decaying non-transferable reputation, rewards only to actual performers	CommunityCredit, ProofOfParticipation, Reputation

Table 1: How the design and its implementation answer the research question.

6. Limitations and future work

Minting the citizenship token still requires an attestation of residency and in the current design that authority rests with the municipality, which could refuse or delay it. Returning control of the attestation to the citizen through the EUDI/IT-Wallet framework would close this gap. Sybil resistance is not enforced on-chain. Anchoring participation to a verifiable on-chain identity presupposes that every resident holds a wallet, a digital identity and a minimum of digital literacy. The model risks skewing participation toward the digitally capable and reproducing the under-representation it sets out to correct. Gasless voting, off-chain communication channels and assisted onboarding lower that barrier without removing it. Finally, one instance governs one neighborhood, but a city is made of many: coordination among instances, and decisions whose scope exceeds a single neighborhood, lie outside the present design.

Future work comprises completing the infrastructural layer, replacing the mock jury with real sortition, adding liquid-democracy delegation in the private flow, building an accessible web dApp with assisted onboarding and validating the model in a real neighborhood.